

Temperature and CO2

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Reading the data

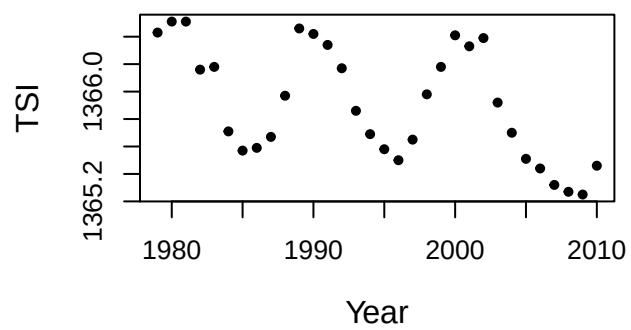
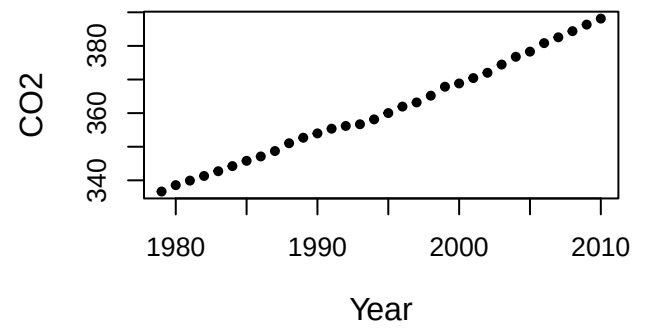
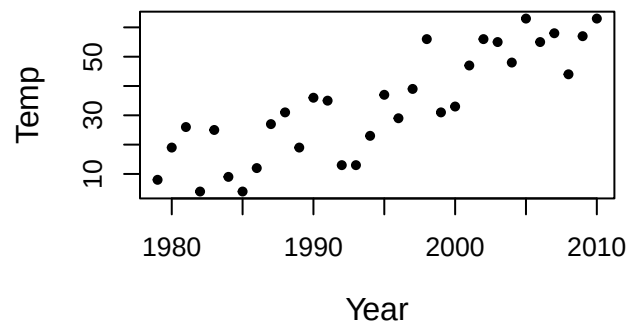
```
a<-read.table("temp_co2_data.txt",header = TRUE)
attach(a)
```

```
## The following object is masked from package:datasets:
##
##      CO2
```

Exploratory analysis

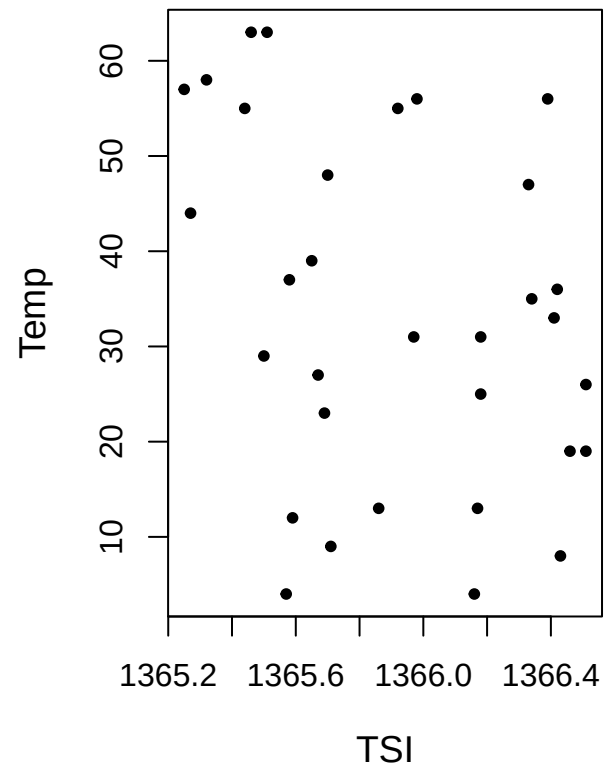
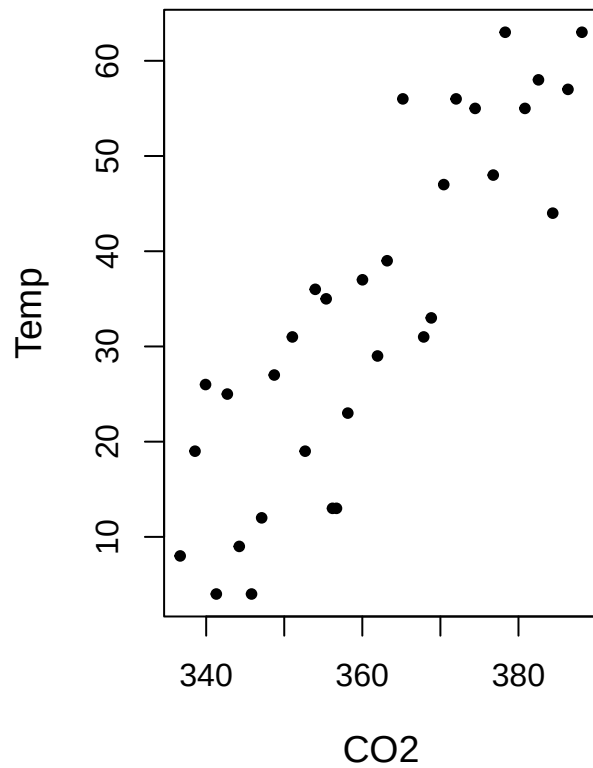
Evolution

```
par(mfrow=c(2,2))
plot(Temp~Year,pch=20,cex.lab=1.2)
plot(CO2~Year,pch=20,cex.lab=1.2)
plot(TSI~Year,pch=20,cex.lab=1.2)
par(mfrow=c(1,1))
```



Scatterplot

```
par(mfrow=c(1,2))
plot(Temp~CO2,pch=20,cex.lab=1.2)
plot(Temp~TSI,pch=20,cex.lab=1.2)
```



```
par(mfrow=c(1,1))
```

Correlation

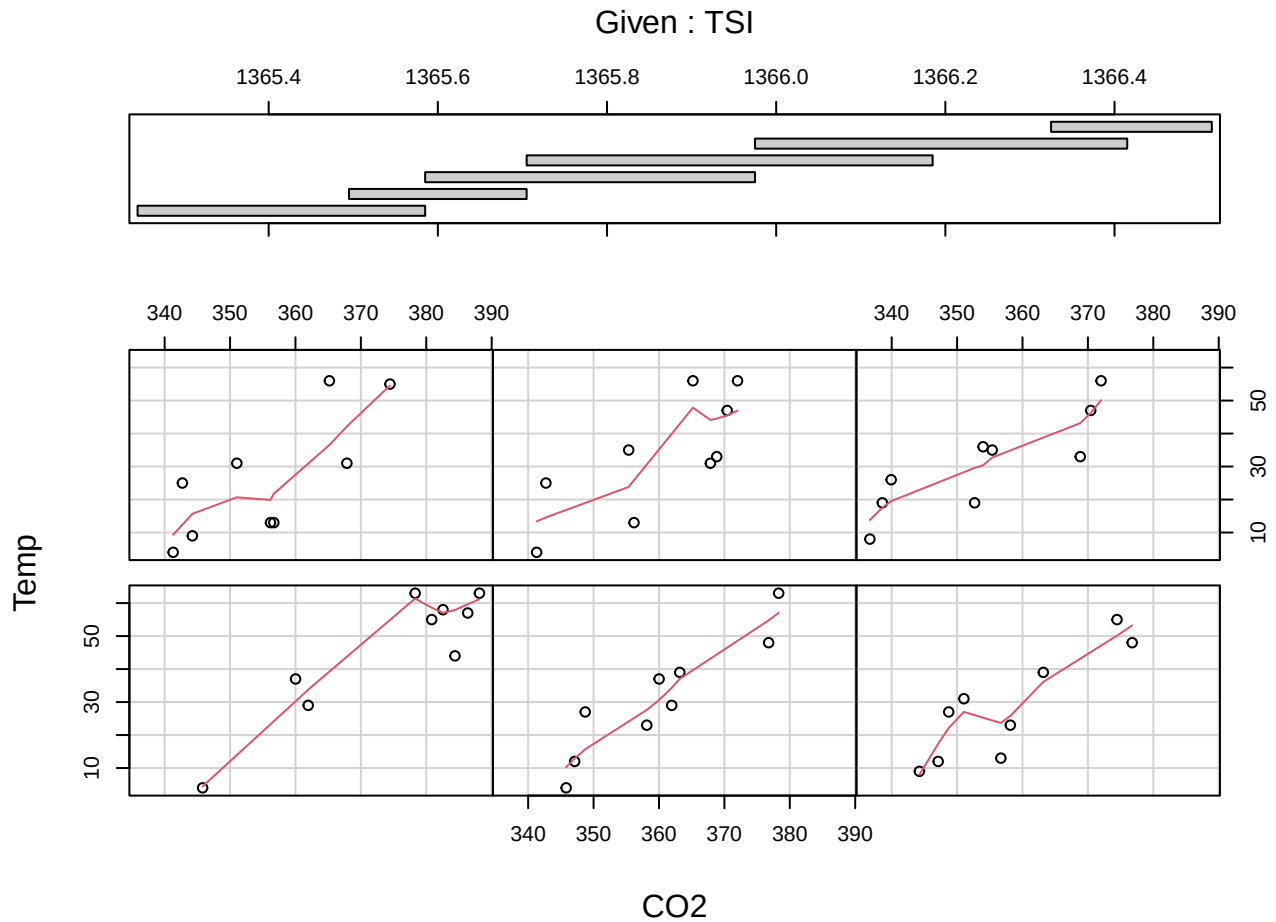
```
cor(cbind(Temp,CO2,TSI))
```

```
##           Temp           CO2           TSI
## Temp  1.0000000  0.8609274 -0.3403420
## CO2   0.8609274  1.0000000 -0.5428219
## TSI  -0.3403420 -0.5428219  1.0000000
```

```
cor(cbind(Temp,CO2,TSI),method = "spearman")
```

```
##           Temp           CO2           TSI
## Temp  1.0000000  0.8612565 -0.3787949
## CO2   0.8612565  1.0000000 -0.5739688
## TSI  -0.3787949 -0.5739688  1.0000000
```

```
coplot(Temp~CO2|TSI,panel = panel.smooth)
```



Significance of a linear model

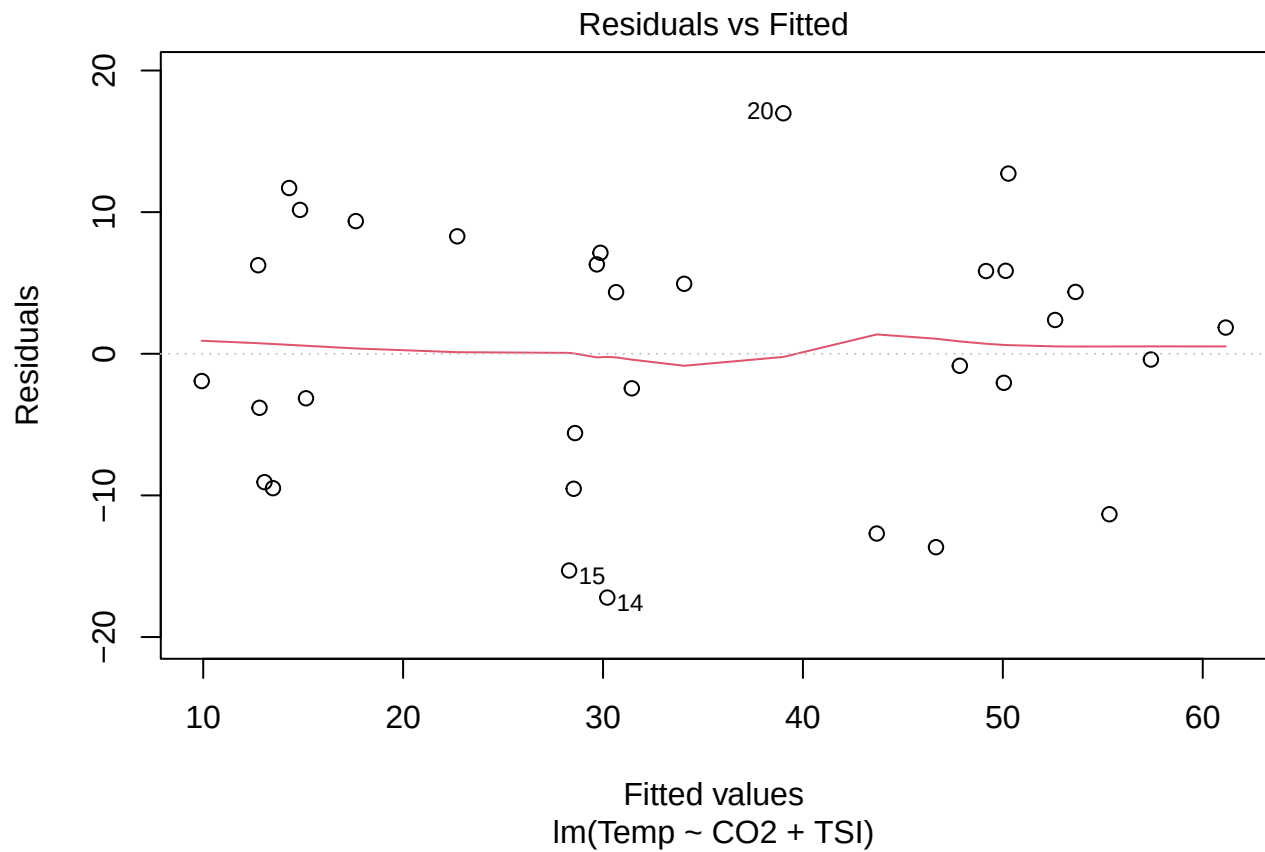
```
fit<-lm(Temp~CO2+TSI)
summary(fit)
```

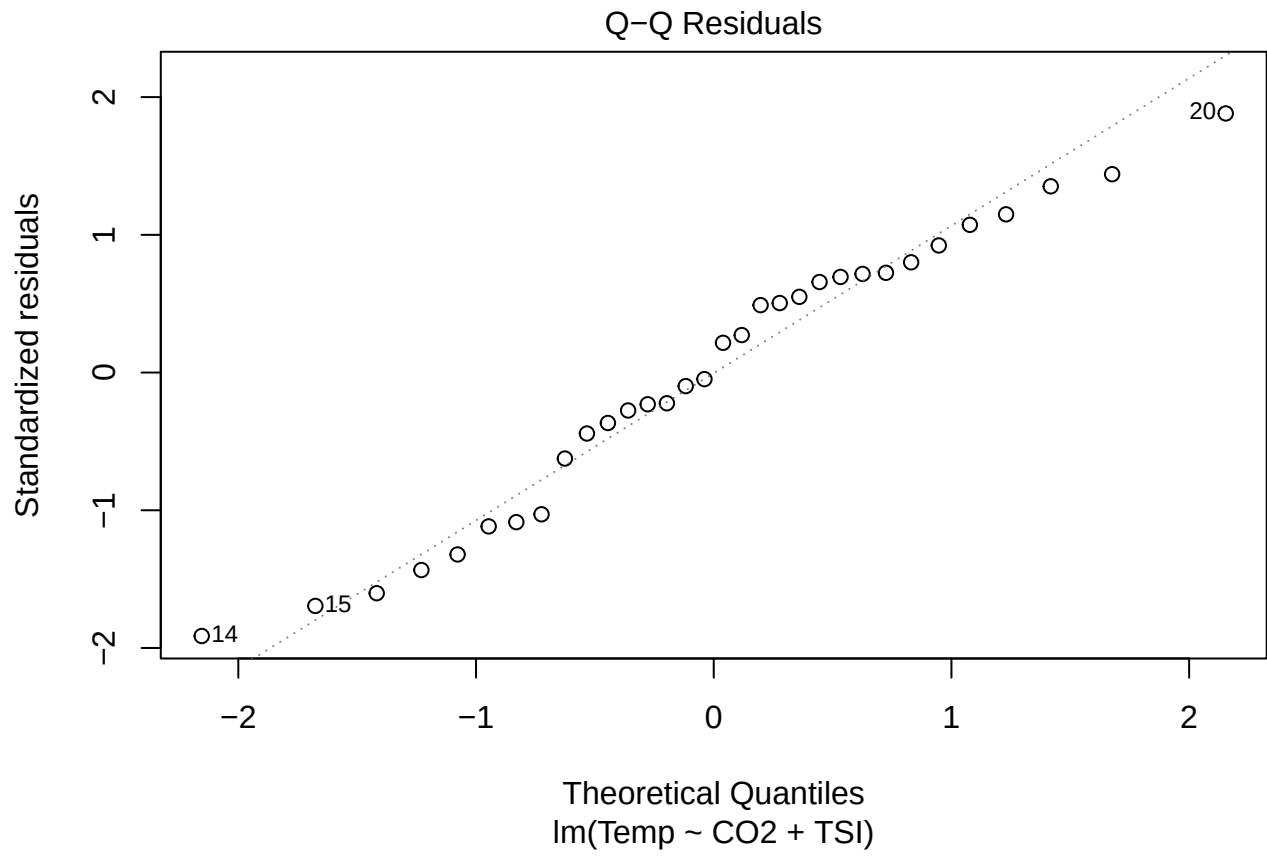
```
##
## Call:
## lm(formula = Temp ~ CO2 + TSI)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -17.2133  -6.4651   0.7219   6.2659  16.9803
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.137e+04  6.583e+03  -1.728  0.0947 .
## CO2          1.147e+00  1.285e-01   8.926 8.13e-10 ***
## TSI          8.047e+00  4.801e+00   1.676  0.1044
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.204 on 29 degrees of freedom
## Multiple R-squared:  0.7641, Adjusted R-squared:  0.7478
```

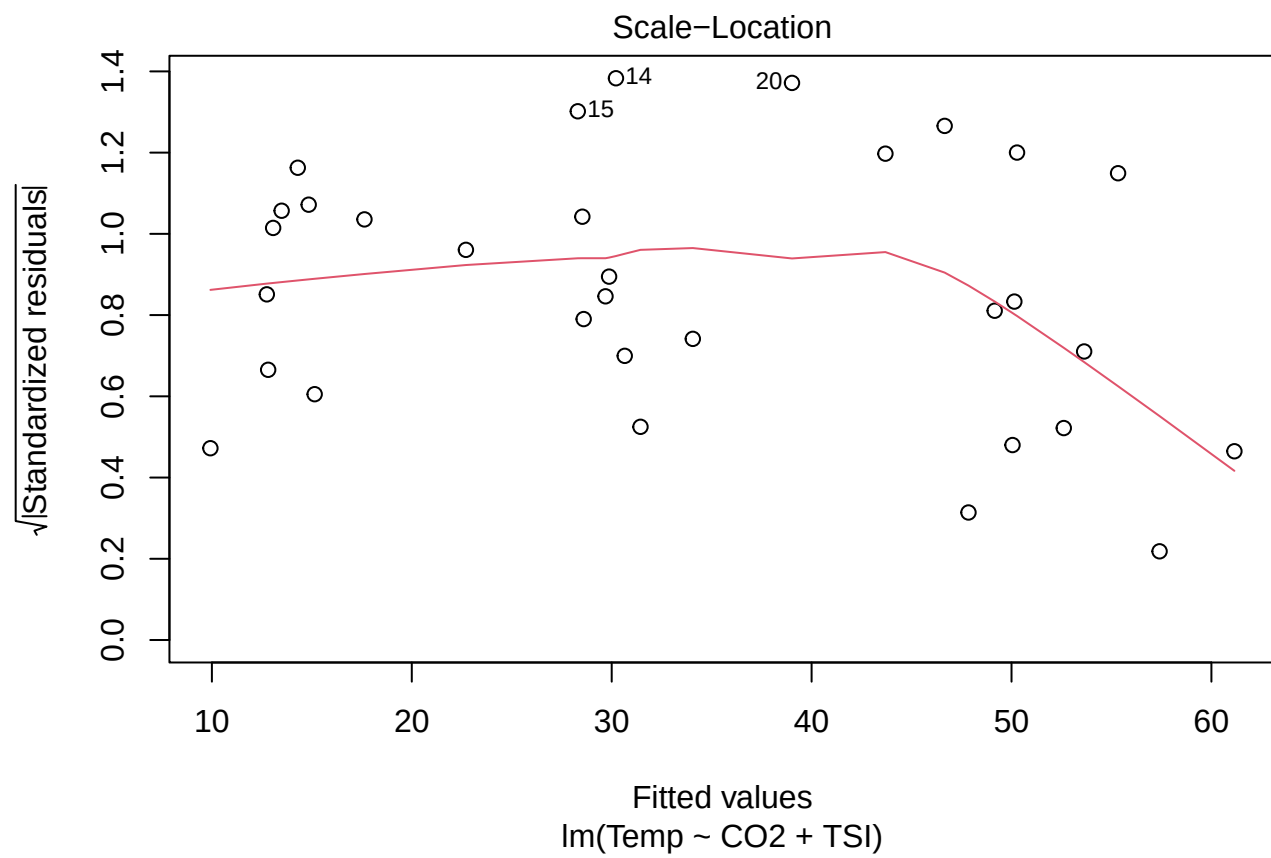
F-statistic: 46.96 on 2 and 29 DF, p-value: 8.047e-10

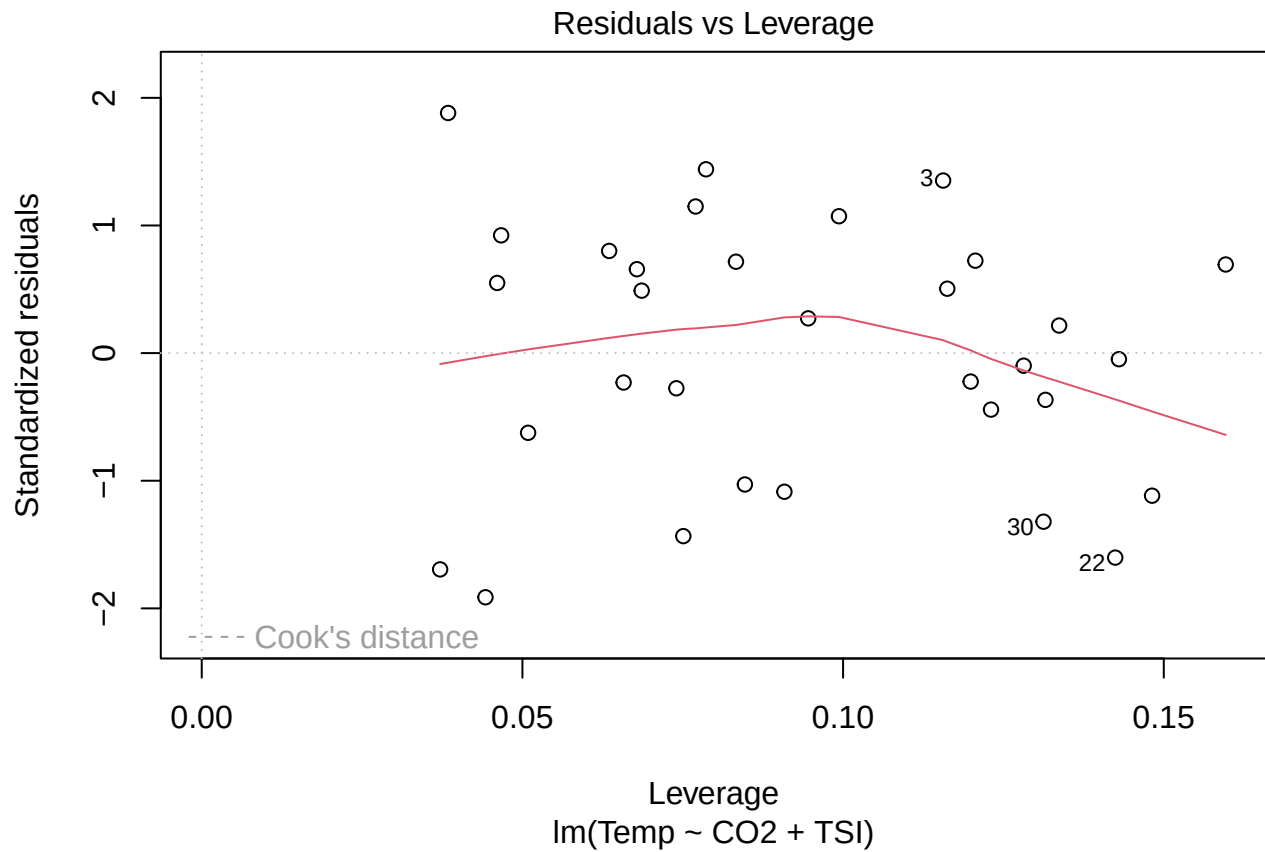
Check of residuals of the first model

```
plot(fit)
```









Confidence interval with different levels

```
confint(fit,level = 0.95)
```

```
##                2.5 %      97.5 %
## (Intercept) -24835.364959 2090.236067
## CO2          0.884273    1.409948
## TSI          -1.770948    17.865870
```

```
confint(fit,level = 0.99)
```

```
##                0.5 %      99.5 %
## (Intercept) -2.951658e+04 6771.44872
## CO2          7.928806e-01    1.50134
## TSI          -5.184953e+00    21.27987
```

```
confint(fit,level = 1)
```

```
##                0 % 100 %
## (Intercept) -Inf    Inf
## CO2          -Inf    Inf
## TSI          -Inf    Inf
```

```
confint(fit,level = 0.90)
```

```
##                5 %      95 %
## (Intercept) -2.255713e+04 -187.997328
## CO2          9.287515e-01    1.365469
```



```
## TSI          -1.094343e-01   16.204357
confint(fit,level = 0)

##              50 %          50 %
## (Intercept) -11372.564446 -11372.564446
## CO2          1.147110      1.147110
## TSI          8.047461      8.047461
```

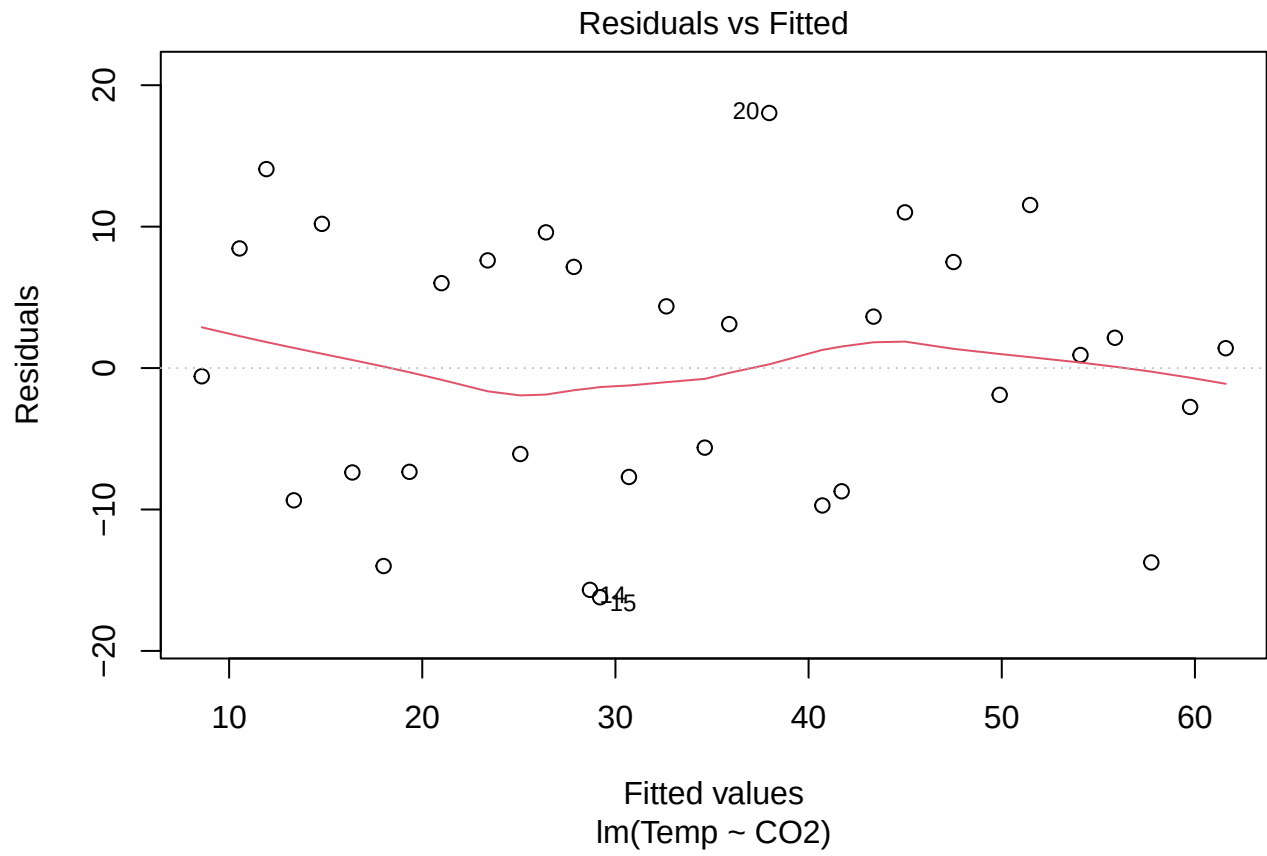
A reduced model

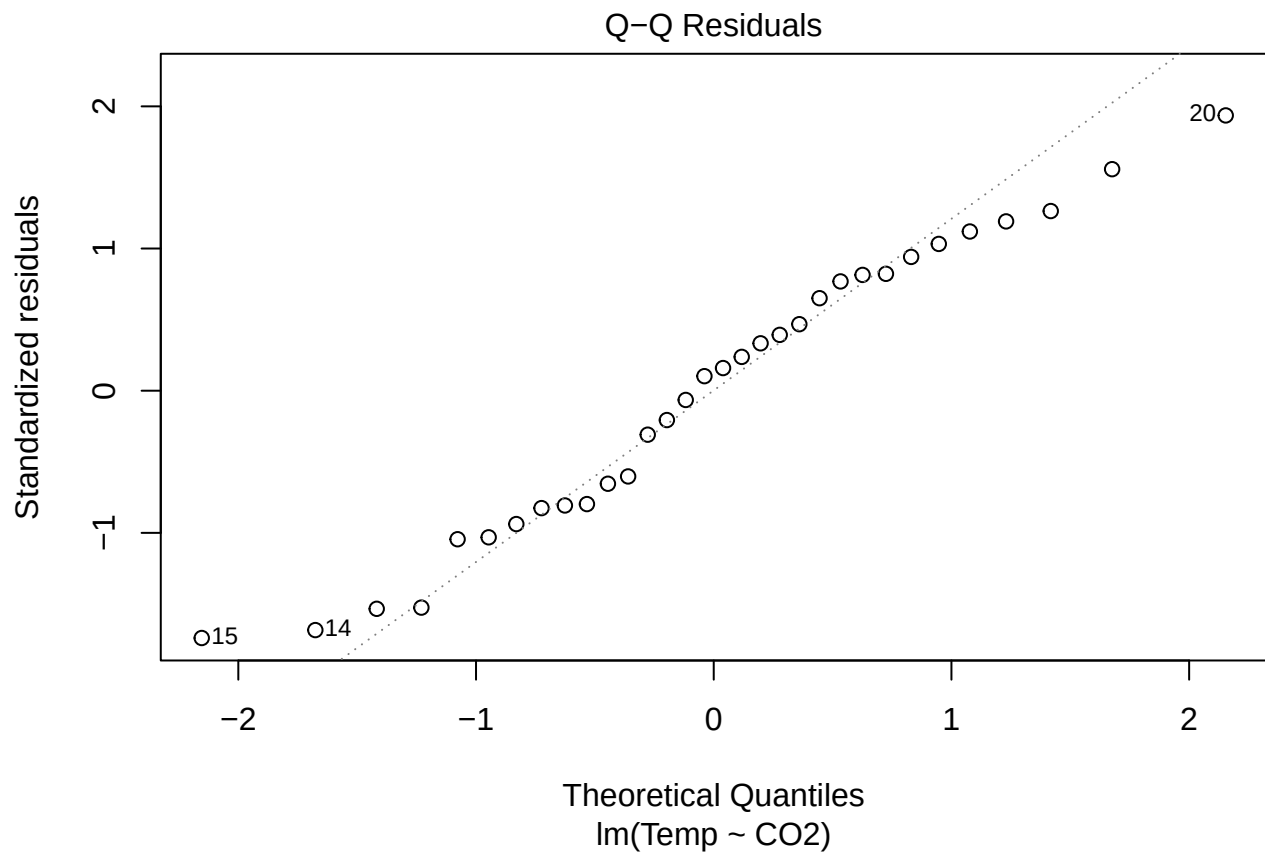
```
fit2<-lm(Temp~CO2)
summary(fit2)

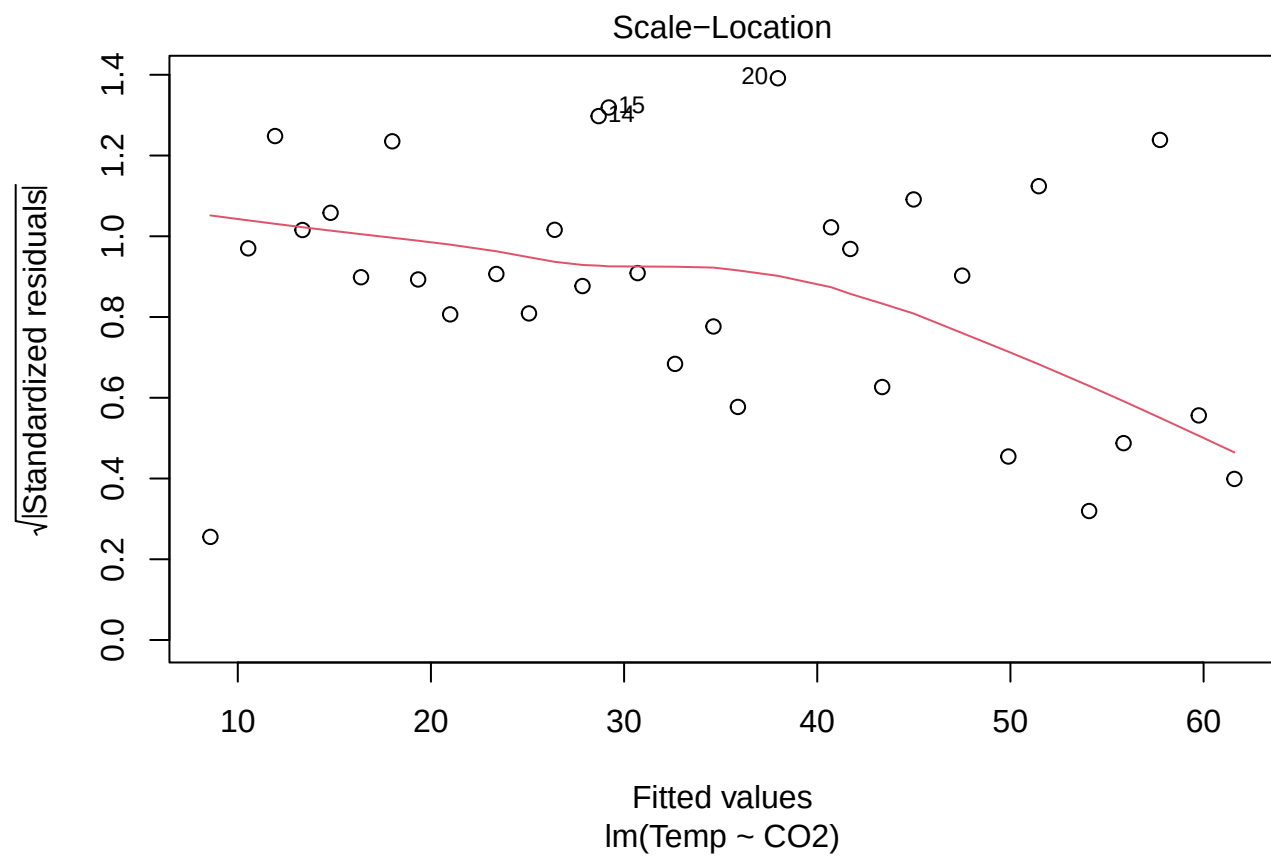
##
## Call:
## lm(formula = Temp ~ CO2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -16.206   -7.461    1.165    7.527   18.037
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -338.2450     40.1505  -8.424 2.11e-09 ***
## CO2          1.0302      0.1111    9.269 2.60e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.477 on 30 degrees of freedom
## Multiple R-squared:  0.7412, Adjusted R-squared:  0.7326
## F-statistic: 85.92 on 1 and 30 DF,  p-value: 2.599e-10
```

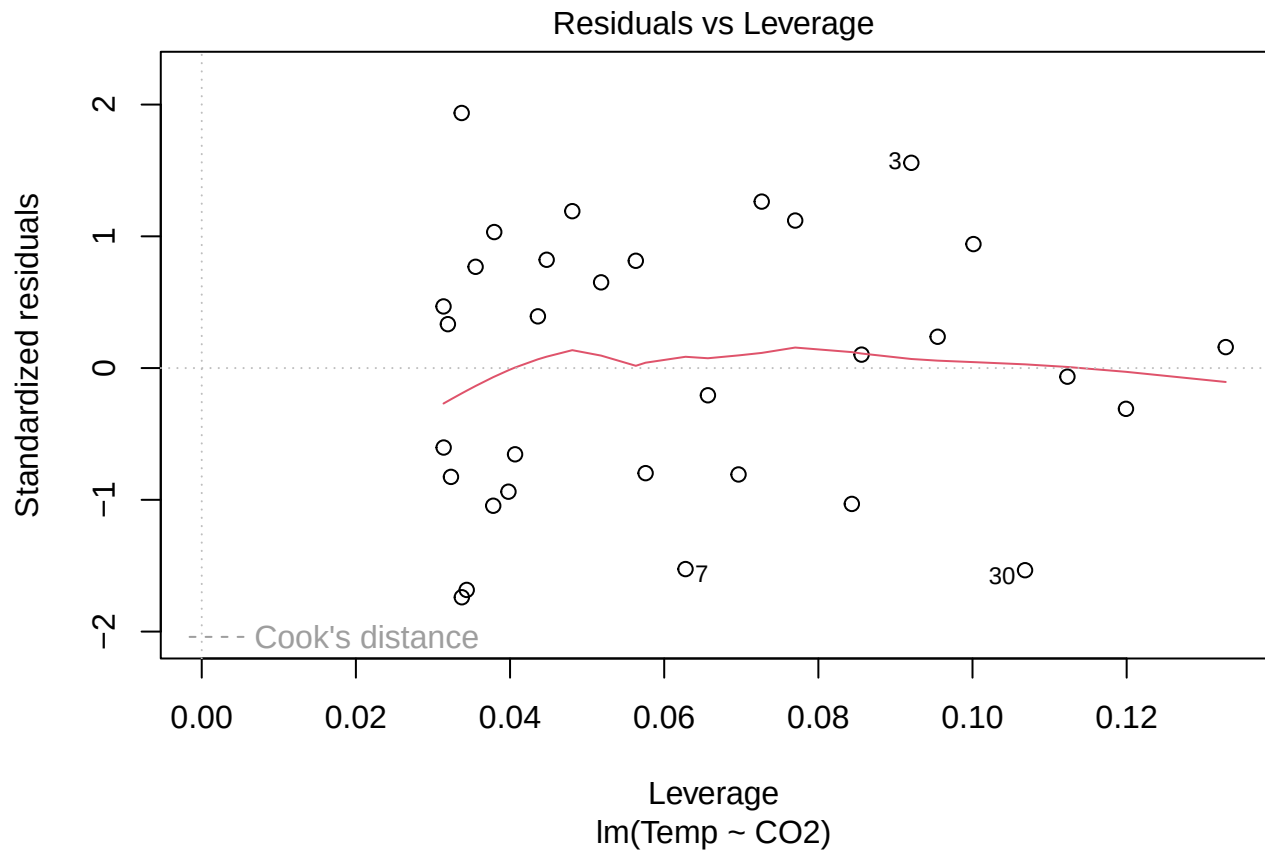
Check residuals of the second model

```
plot(fit2)
```

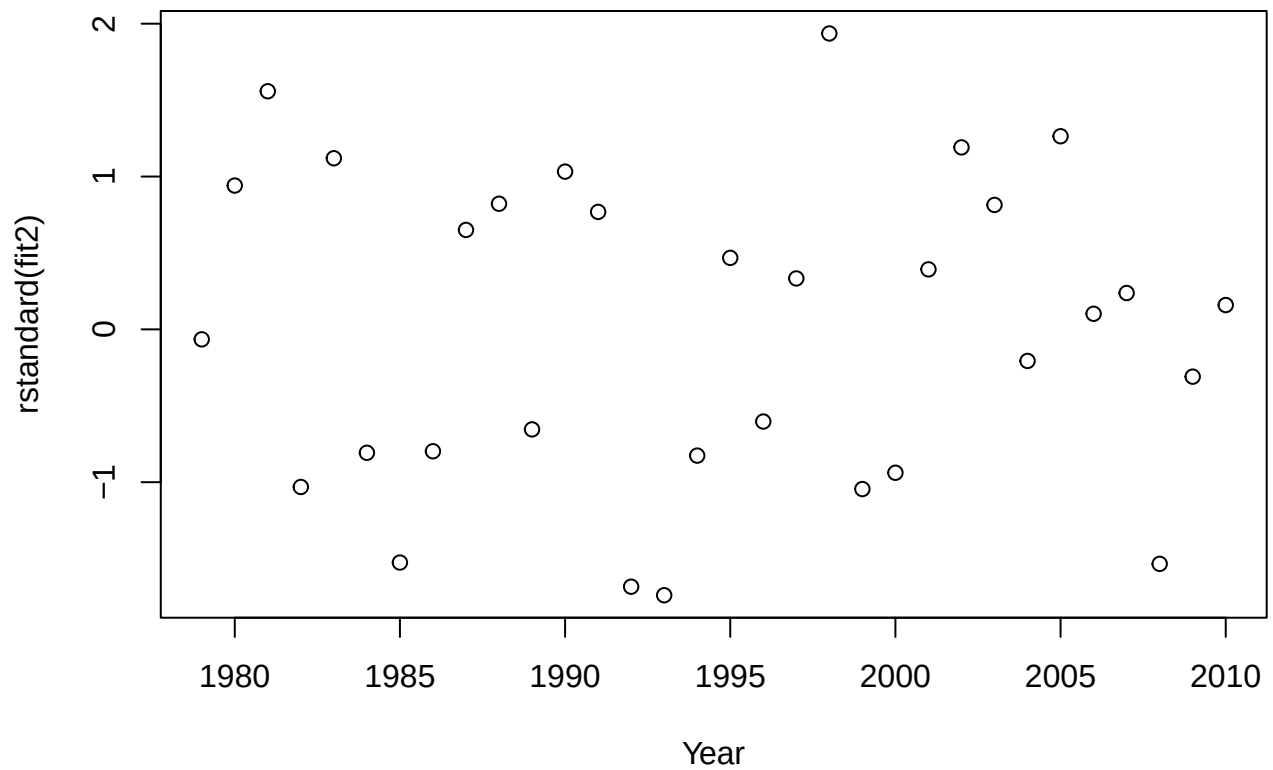








```
plot(rstandard(fit2)~Year)
```



Confidence interval for the mean $E(Y)$ at observed values of covariates

```
predict(fit2,interval="confidence",level=0.95)
```

##	fit	lwr	upr
## 1	8.582496	2.095971	15.06902
## 2	10.539820	4.415448	16.66419
## 3	11.930551	6.057821	17.80328
## 4	13.352186	7.731067	18.97330
## 5	14.804726	9.433992	20.17546
## 6	16.380887	11.273038	21.48874
## 7	17.998255	13.149114	22.84740
## 8	19.337477	14.692755	23.98220
## 9	20.996052	16.590228	25.40188
## 10	23.386047	19.291496	27.48060
## 11	25.075527	21.173024	28.97803
## 12	26.404447	22.634010	30.17488
## 13	27.846686	24.198333	31.49504
## 14	28.681124	25.092436	32.26981
## 15	29.206511	25.651024	32.76200
## 16	30.700258	27.219808	34.18071
## 17	32.636979	29.208927	36.06503
## 18	34.625208	31.196105	38.05431
## 19	35.892318	32.433476	39.35116
## 20	37.962961	34.408552	41.51737
## 21	40.713516	36.949490	44.47754
## 22	41.712782	37.851808	45.57376
## 23	43.361055	39.318988	47.40312
## 24	44.988724	40.744853	49.23260
## 25	47.502340	42.909077	52.09560
## 26	49.892336	44.932218	54.85245
## 27	51.468497	46.251455	56.68554
## 28	54.074829	48.411749	59.73791
## 29	55.857024	49.876341	61.83771
## 30	57.742236	51.416412	64.06806
## 31	59.751069	53.048696	66.45344
## 32	61.595074	54.540293	68.64985

```
predict(fit2,interval="confidence",level=0.99)
```

##	fit	lwr	upr
## 1	8.582496	-0.1518504	17.31684
## 2	10.539820	2.2931252	18.78652
## 3	11.930551	4.0227017	19.83840
## 4	13.352186	5.7831410	20.92123
## 5	14.804726	7.5728326	22.03662
## 6	16.380887	9.5029790	23.25880
## 7	17.998255	11.4687061	24.52780
## 8	19.337477	13.0831870	25.59177
## 9	20.996052	15.0634468	26.92866
## 10	23.386047	17.8725824	28.89951
## 11	25.075527	19.8206621	30.33039
## 12	26.404447	21.3274129	31.48148

```
## 13 27.846686 22.9340429 32.75933
## 14 28.681124 23.8488226 33.51343
## 15 29.206511 24.4189156 33.99411
## 16 30.700258 26.0137027 35.38681
## 17 32.636979 28.0209797 37.25298
## 18 34.625208 30.0077933 39.24262
## 19 35.892318 31.2348594 40.54978
## 20 37.962961 33.1768168 42.74911
## 21 40.713516 35.6451146 45.78192
## 22 41.712782 36.5138370 46.91173
## 23 43.361055 37.9182623 48.80385
## 24 44.988724 39.2741945 50.70325
## 25 47.502340 41.3173412 53.68734
## 26 49.892336 43.2133535 56.57132
## 27 51.468497 44.4435564 58.49344
## 28 54.074829 46.4492822 61.70038
## 29 55.857024 47.8038131 63.91023
## 30 57.742236 49.2242792 66.26019
## 31 59.751069 50.7260747 68.77606
## 32 61.595074 52.0955501 71.09460
```

Confidence interval for the mean $E(Y)$ at specific values

```
CO2.new<-data.frame(CO2=c(370,395,420,0))
predict(fit2,CO2.new,interval="confidence",level=0.95)
```

```
##          fit          lwr          upr
## 1  42.91808  38.92712  46.90904
## 2  68.67235  60.22001  77.12468
## 3  94.42661  80.59352 108.25970
## 4 -338.24503 -420.24337 -256.24669
```

```
predict(fit2,CO2.new,interval="confidence",level=0.99)
```

```
##          fit          lwr          upr
## 1  42.91808  37.54411  48.29206
## 2  68.67235  57.29096  80.05373
## 3  94.42661  75.79985 113.05337
## 4 -338.24503 -448.65883 -227.83123
```

```
predict(fit2,CO2.new,interval="confidence",level=0.99999)
```

```
##          fit          lwr          upr
## 1  42.91808  32.56202  53.27414
## 2  68.67235  46.73954  90.60515
## 3  94.42661  58.53142 130.32180
## 4 -338.24503 -551.02085 -125.46921
```

Prediction and prediction interval

```
CO2.new<-data.frame(CO2=c(370,395))
predict(fit2,CO2.new,interval="confidence",level=0.95)
```

```
##          fit          lwr          upr
```

```
## 1 42.91808 38.92712 46.90904
## 2 68.67235 60.22001 77.12468
```

```
predict(fit2,C02.new,interval="prediction",level=0.95)
```

```
##          fit          lwr          upr
## 1 42.91808 23.15562 62.68055
## 2 68.67235 47.55199 89.79270
```

```
plot(C02,Temp,ylim=c(-30,100))
abline(fit2)
C02.new<-data.frame(C02=seq(330,450,by=1))
pred<-predict(fit2,C02.new, interval="confidence",level=0.95)
segments(x0=C02.new$C02,y0=pred[,2],x1=C02.new$C02,y1=pred[,3],lwd=2,col="red")
pred<-predict(fit2,C02.new, interval="prediction",level=0.95)
segments(x0=C02.new$C02-0.5,y0=pred[,2],x1=C02.new$C02+0.5,y1=pred[,3],lwd=2,col="blue")
```

